

Assignment 6: Data Normalization

The purpose of this assignment is to get practice normalizing data using the first three normal forms. I recommend working in Microsoft Word or Excel (or similar program, like Google Docs). You will have to convert your work to PDF form for hand in, so make sure your chosen is able to do that well.

Task 1: Employees

Based on a lesson by Tomoko Okochi, instructor at Langara College

Consider the following table of data for a workplace:

Emp. ID	Employee Name	Email	Department	Job Type	Hourly Rate	Manager ID	Projects
101	Alice Smith	alice@email.com	HR	Recruiter	25	202	Employee Training, Hiring
202	Bob Miller	bob@email.com	HR	Manager	40	202	HR Strategy
103	Charlie Green	charlie@email.com	IT	Developer	35	203	Software Development, Maintenance
104	Garth White	garth@email.com	IT	Developer	35	203	Software Development, Testing
105	Eric Johnson	eric@email.com	IT	Tester	30	203	Testing
106	Fiona Adams	fiona@email.com	Sales	Sales Rep	28	204	Customer Outreach, Sales Reports

- [1 mark] Write the above relation as a single line using the shorthand notation from our lessons and slides, with the **Primary Key** underlined.
- [2 marks] convert this data to **First Normal Form** by moving the multi-valued attribute into its own table. Figure out what the **Primary Key** for the new table should be. Write the relation shorthand notation AND the full tables populated with data. Remember to clearly indicate the Primary Keys by underlining their column/attribute names.
- [1 mark] What is the **Foreign Key** in the new table?
- [1 mark] What, if anything, needs to be done to get this data into **Second Normal Form**? Explain your reasoning.
- [1 mark] Determine all **transitive dependencies** in the data, along with their **determinants**, and write them in the form **(A,B, ...) ->> (C, D, ...)** where the left-hand side is the determinant.

- f) [2 marks] Convert the data to **Third Normal Form**, writing the shorthand relation notation for each table AND the full tables themselves, populated with data. PKs should be clearly underlined.

Review your work thoroughly, checking the following:

- All Primary Key columns should have unique values in EVERY row;
- No data should be found in more than one table UNLESS it belongs to a Primary or Foreign key;
- You should be able to trace a path from any column, in any table, to any other column in any other table by following connections between foreign and primary keys.

Task 2: Movies

Consider the following table of data about movies:

Movie Productions

movie_id	title	prod_company	prod_company_city	actor_id	actor_award
1	Tenet	Syncopy	London	1	none
1	Tenet	Syncopy	London	2	best supporting actor
1	Tenet	Syncopy	Los Angeles	1	none
1	Tenet	Syncopy	Los Angeles	2	best supporting actor
2	Mickey 17	Warner Bros.	Burbank	2	best actor
2	Mickey 17	Plan B	Los Angeles	2	best actor
2	Mickey 17	Offscreen	Los Angeles	2	best actor
2	Mickey 17	Warner Bros.	Burbank	3	none
2	Mickey 17	Plan B	Los Angeles	3	none
2	Mickey 17	Offscreen	Los Angeles	3	none
3	Dunkirk	Syncopy	London	4	none
3	Dunkirk	Syncopy	Los Angeles	4	none

Note that an actor can only win one award for any given movie.

- [1 mark] What is a minimal composite primary key of this table? Convert the table to relation notation with the PK columns underlined.
- [2 marks] Which non-key columns are NOT functionally dependent on the entire primary key? What are their determinants? (Which columns in the PK are they functionally determined by?)
- [1 mark] Using your answer for part b), convert this data into second normal form and write the relations with their keys. (You don't have to write the tables, although you can if you find it helpful in your process.)

Perform the same checks that you did for Task 1 to make sure your normalized data model makes sense.

Total marks: **12**

Hand In

Output your work as a PDF file named **a6-firstname-lastname**. Zip your file and hand it in to the appropriate folder on Brightspace.

As usual, up to -25% could be deducted for improper hand in, disorganized files, etc. Ask me if in doubt.